# Learning-rate schedules, batch size, and optimizer memory

This companion expands notes §§6.3–6.9 and blog §§6.3–6.9. Width transfer, token-horizon tuning and schedule choice are different interventions. Check each under stated conditions.

## Start with units and parameter groups

Token batch is $B = b\ell$ for $b$ fully populated sequences of length $\ell$; with masks, distinguish processed and scored tokens. The approximate update count is $S = D/B$. µP can keep a base LR stable across width while changing actual matrix-group LRs. A raw global LR from a conventional recipe is not automatically a µP base LR. Tensor Programs V, Appendix B.1; architecture evidence.

## Decay and optimizer-memory timescales

With decoupled decay, the decay-only update multiplies a parameter by $1 - \eta_s\lambda$. At small products,

$$\prod_{s=1}^{S}(1 - \eta_s\lambda) \approx \exp[-\lambda \sum_{s} \eta_s].$$

For constant LR this leads to $\tau = B/(\eta\lambda D)$, a dimensionless fraction-of-training proxy. Scheduled runs require the full LR shape and matching parameter-group conventions.

Power Lines v2 finds $\tau_* \approx 1.084(D/N)^{-0.527}$, so its cross-TPP optimum is not constant. At fixed $N$, $D$, $\eta$, holding $\tau$ fixed motivates $\lambda \propto B$ in its supported regime. Moving to a new TPP requires the fitted $\tau_*$ or new tuning. Bergsma et al., v2, §2/Fig. 1.

Adam's momentum coefficients are measured in steps. A coefficient $\beta$ has token e-folding span $-B/\ln\beta$. To preserve that span when batch changes, the derived relation is $\beta_{\text{new}} = \beta_{\text{old}}^{B_{\text{new}}/B_{\text{old}}}$. This is an accounting identity under a constant-batch exponential-memory model; whether preserving the span is optimal is an empirical question.

## Compare schedules at comparable endpoints

Cosine ties decay to a chosen horizon. WSD provides a stable trunk that can fork into multiple decay branches. A branch comparison must record its starting checkpoint, tokens added, decay shape, final LR and mixture. Stable-phase loss is not the same quantity as annealed endpoint loss. Tune candidate schedules comparably. Fantastic Optimizers; Tissue et al., loss with annealing.

In a branch budget, if shared trunk cost is $C_0$ and branches cost $C_j$, actual training cost is $C_0 + \sum_j C_j$. Summing full end-to-end costs for every branch double-counts the prefix.

## Two clocks in an actual production run

Nano's report uses a 25T-token horizon, starts its final 5T-token LR decay around 80%, and switches to the quality-focused mixture at 94%. These boundaries differ. Reproducing the data curriculum while moving decay earlier is a different experiment. Super and Ultra use different phase boundaries again. Nano §§2.3–2.4; full comparison.

## A small calibration experiment

At fixed shape, compare two token horizons and several token batches. First tune LR and decay on the smaller configuration; then test the transferred region at the longer horizon. Include a fully annealed comparison for schedule finalists, preserve branch lineage and confirm the selected setting with a fresh seed. This is our proposed design, not a published universally sufficient run count.

**Required artifact:** a schedule function, batch/optimizer-memory units, matrix-group hyperparameters, full endpoint definition and measured LR/batch/decay response.

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